Session 01(AIML) - Assignment Question

Name of Student : Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College name: Zeal Polytechnic , Narhe

Branch Name: AIML

Q1.

PROGRAM:

```python
#Q1
#personal information

Full_name = "Shruti Dnyaneshwar Darwatkar" #string
Age = 18                                   #integer
Height_m = 4.9                             #float
is_student = True                          #boolean

print("Name:", Full_name)
print("age:", Age)
print("height:",Height_m)
print("Student:",is_student)

print(type(Full_name))
print(type(Age))
print(type(Height_m))
print(type(is_student))
```

OUTPUT:

```
Name: Shruti Dnyaneshwar Darwatkar
age: 18
height: 4.9
Student: True
<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>
```

Q2.

PROGRAM:

```python
#Q2

name = "Shruti dnyaneshwar darwatkar"      # str
age = 18                                   # int
height = 4.9                               # float
is_student = True                          # bool
print(name, type(name))
print(age, type(age))
print(height, type(height))
print(is_student, type(is_student))
age_float = float(age)
height_int = int(height)
print("Integer to Float:", age_float, type(age_float))
print("Float to Integer:", height_int, type(height_int))
# Explanation:
# float(age) converts integer 18 to 18.0
# int(height) converts 4.9 to 4 (decimal part is removed)
```

OUTPUT:

```
Shruti dnyaneshwar darwatkar <class 'str'>
18 <class 'int'>
4.9 <class 'float'>
True <class 'bool'>
Integer to Float: 18.0 <class 'float'>
Float to Integer: 4 <class 'int'>
```

Q3.

PROGRAM:

```
#Q3

#Arithmetic operation
a=15
b=4
print("Addition =",a + b)
print("Subtraction =",a - b)
print("Multiplication =",a * b)
print("Division =",a / b)
print("Floor Division =",a // b)
print("Modulo =",a % b)
print("Exponentiation =",a ** b)

#  / gives the exact decimal result
# // gives the integer (floor) reuslt
print("15/4 = ",15/4)
print("15//4 =" ,15//4)
```

OUTPUT:

```
Addition = 19
Subtraction = 11
Multiplication = 60
Division = 3.75
Floor Division = 3
Modulo = 3
Exponentiation = 50625
15/4 =  3.75
15//4 = 3
```

Q4.

PROGRAM:

```
#Q4
x = 25
y = 30
z = 25

# 1. Is x greater than y?
print("Is x greater than y?", x > y)
# 2. Is x equal to z?
print("Is x equal to z?", x == z)
# 3. Is x less than or equal to y AND y is not equal to z?
print("Is x <= y AND y != z?", (x <= y) and (y != z))
# 4. Is x greater than y OR x equal to z?
print("Is x > y OR x == z?", (x > y) or (x == z))
```

OUTPUT:

```
Is x greater than y? False
Is x equal to z? True
Is x <= y AND y != z? True
Is x > y OR x == z? True
```

Q5.

PROGRAM:

```
#Q5
name = input("Enter your name: ")
# Convert input to integer because input() gives data as text
birth_year = int(input("Enter your birth year: "))
age = 2026 - birth_year
print("Name:", name)
print("Age:", age)
```

OUTPUT:

```
Enter your name:  shruti
Enter your birth year:  2008
Name: shruti
Age: 18
```

Q6.

PROGRAM:

```
#Q6

# Take height and weight from user
height = float(input("Enter height in meters: "))
weight = float(input("Enter weight in kg: "))
# Calculate BMI
bmi = weight / (height ** 2)
# Print BMI up to 2 decimal places
print("BMI =", round(bmi, 2))
```

OUTPUT:

```
Enter height in meters:  0.124
Enter weight in kg:  43
BMI = 2796.57
```

Q7.

PROGRAM:

```
#Q7s

"""
This program calculates the area and perimeter of a rectangle.
The formulas are useful in many real-life measurement problems.
"""
# Taking input as float allows decimal values
length = float(input("Enter length: "))
width = float(input("Enter width: "))

# Area helps find the space covered by the rectangle
area = length * width

# Perimeter helps find the total boundary length
perimeter = 2 * (length + width)

# Displaying results clearly improves readability
print("Area =", area)

# Showing perimeter separately avoids confusion
print("Perimeter =", perimeter)
```

OUTPUT:

```
Enter length:  6.6
Enter width:  4.5
Area = 29.7
Perimeter = 22.2
```

Q8.

PROGRAM:

```python
#Q8

fruits = ["mango", "apple", "banana", "cherry"]
score = 50
score += 25
print("Updated score =", score)
print("Is apple in fruits?", "apple" in fruits)
print("Is grape not in fruits?", "grape" not in fruits)
```

OUTPUT:

```
Updated score = 75
Is apple in fruits? True
Is grape not in fruits? True
```

Q9.

PROGRAM:

```
#Q9
# Taking input
name = input("Enter your name: ")
age = int(input("Enter your age: "))
city = input("Enter your city: ")

# Marks in 3 subjects
marks1 = float(input("Enter marks in Subject 1: "))
marks2 = float(input("Enter marks in Subject 2: "))
marks3 = float(input("Enter marks in Subject 3: "))

# Calculating total marks and percentage
total_marks = marks1 + marks2 + marks3
percentage = total_marks / 3

# Displaying student profile
print("\n===== STUDENT PROFILE =====")
print("Name:", name)
print("Age:", age)
print("City:", city)
print("Subject 1:", marks1)
print("Subject 2:", marks2)
print("Subject 3:", marks3)
print("Total Marks:", total_marks)
print("Percentage:", percentage, "%")
```

OUTPUT:

```
Enter your name:  shruti
Enter your age:  18
Enter your city:  pune
Enter marks in Subject 1:  90
Enter marks in Subject 2:  89
Enter marks in Subject 3:  78

===== STUDENT PROFILE =====
Name: shruti
Age: 18
City: pune
Subject 1: 90.0
Subject 2: 89.0
Subject 3: 78.0
Total Marks: 257.0
Percentage: 85.66666666666667 %
```

Q10.

PROGRAM:

```
#Q10
# Fixed variable name (use same case everywhere)
name = "Alice"
# Added '=' for assignment
age = 20
# Added quotes because Amsterdam is a string
city = "Amsterdam"
# Print name
print("My name is " + name)
# Convert age to string while printing
print("I am", age, "years old")
# Print city
print("I live in", city)
# Check if age is greater than 18
print("Adult:", age > 18)
```

OUTPUT:

```
My name is Alice
I am 20 years old
I live in Amsterdam
Adult: True
```